

**PROGRAMME: BCA (Bachelor of Computer Applications)
SEMESTER – I**

Teaching-Learning & Evaluation Plan

Course Information: Core Course

Course Code: **23BCA1C03**

Course Title: **Fundamentals of Mathematics**

Credits Units: **3+1**

Contact Hours: **45**

L-T-P: **3-0-0**

CA: UE Weightage - **50:50**

Pass Marks (CA & UE) – **20:20**

Aggregate Pass Marks: **40**

UE Question Paper Marks: **50**

Special Examination Fees: **NA**

Prerequisite (if any): **NA**

Course Facilitator (s): Ayesha, Lavanya.D

Programme Outcomes (POs)

At the end of the programme, students will be able to

PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand, and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well-informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11	Individual & Team Work: Ability to work as a member or leader in diverse teams in multidisciplinary environment.

PO12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.
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Program Specific Outcomes (PSO's)

PSO 1	Pertain current knowledge and adapting to emerging applications of Mathematics, Science fundamentals in the field of Computer science and its applications.
PSO 2	Exhibit proficiency in identifying, formulating and analyzing complex problems in the computer environment.
PSO 3	Ability to create, select and apply appropriate modern techniques for solving complex issues.
PSO 4	Explore technical knowledge in diverse areas of Computer Applications and experience a conducive environment in nurturing skills for successful career and higher studies.

Course Objectives (COB's)

COB1: To explain, Describe the concept of Matrices & Determinants and students will learn to understand the concept of Mathematical Logic's.

COB2: To define and understand the concept of Set theory.

COB3: To find out the number of ways in which a given number of objects can be arranged by understanding the concept of permutation and combination.

Course Outcomes (CO's)

At the end of the course, students will be able to

CO1: Explain the concept of Matrices

CO2: Discuss the concept of the system of equations.

CO3: Identify and Analyze statements and prove the concepts of Mathematical logics.

CO4: Analyze Understand and Explain set theory.

CO5: Identify and Implement the concept of Permutation and combination.

Sl. No.	Course Outcome	Skill's	Pos	Bloom's Taxonomy Level	Assessment Tool
1.	CO 1	Understanding	PO-1 PO-2 PO-4 PO-7 PO-8 PO-9 PO-10	L2	Presentation
2.	CO 2	Apply, Analyze and Evaluate	PO-1 PO-2 PO-4 PO-7 PO-8 PO-9 PO-10	L3, L4, L5	Assignment
3.	CO 3	Apply, Analyze and Evaluate	PO- 1, PO-2, PO- 4, PO- 5, PO-6, PO-7, PO-9, PO-10.	L3, L4, L5	Quiz
4.	CO 4	Apply, Analyze	PO- 1, PO-2, PO-3, PO-6, PO-7, PO-8.	L3, L4	Poster Presentation
5.	CO 5	Understanding	PO-1, PO-2, PO-4 PO-5, PO-6, PO-7, PO-9, PO-10	L2	Flip class activity on counting principle

CO-PO/PSO Mapping:

Course Outcome	Blooms Taxonomy Levels	Program Outcomes (PO's) Mapped	Program Specific Outcomes (PSO's)
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		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
														L3	L4	L6	L4
CO1	L2	3	2	-	1	-	-	1	1	1	1	-	-	2	2	2	1
CO2	L4	3	2	-	1	-	-	1	1	1	1	-	-	2	2	2	1
CO3	L3	3	3	-	1	1	2	1	-	2	2	-	-	2	2	1	1
CO4	L4	3	3	1	-	-	1	1	1	-	-	-	-	1	1	1	-
CO5	L4	3	3	-	2	1	2	1	-	2	2	-	-	2	1	2	1

Course Contents:

Module	Details	Contact Hours
I	Matrix Theory : Introduction, matrix definition - types of matrix, operations on matrices. Elementary row and transformation, Inverse of a matrix by elementary transformation, Rank of a matrix, Determination of rank of a matrix by reducing it to echelon form, Linear equations, Homogeneous linear equations, Non – homogenous equations.	9
II	System of equations by Matrix theory: Linear equations, Homogeneous linear equations, Non – homogenous equations. Trivial equations, non trivial equations and their problems.	9
III	Mathematical Logic: Proposition, compound proposition, logical connectives - problems, laws of logics, problems on laws of logic. Open sentences, Compound open sentences, quantifiers, Truth sets, Connectives involving quantifiers, Logical implications.	9
IV	Set Theory : Fundamentals of set operations, Subsets, Counting and Venn Diagrams, Relations, Functions: Plain and one to one, Onto Functions, Function Composition and Inverse Functions, the Pigeonhole Principle - problems.	9
V	Permutations and Combinations : Introduction, The rule of sum and product, Permutations problems, Permutations with Repetitions and Permutations of Objects not all Different, Combinations problems, Combinations with Repetition.	9

Text Books (TB):

TB-1	Discrete and Combinatorial mathematics – Grimaldi and Ramanna – Pearson Education-2017.
TB-2	Discrete Mathematical Structures- Dr. KSC- Sudha Publication-2019.
TB-3	Discrete Mathematical Structures- Dr. DSC- Prism Publication- sixth edition(2019).

Reference Books (RB):

RB-1	Advanced engineering mathematics- B S Grewal - 44th edition.
RB-2	Higher Engineering Mathematics – B V Ramana-2017.

Other Reading (OR):

OR-1	https://www.tutorialspoint.com/discrete_mathematics/index.htm
OR-2	https://www.udemy.com/master-discrete-mathematics/

Relevant Website (RW):

RW-1	https://math.hmc.edu/calculus/hmc-mathematics-calculus-online-tutorials/linear-algebra/matrix-algebra/
RW-2	https://algebra.nipissingu.ca/tutorials/sys_linear_equ.html
RW-3	https://www.tutorialspoint.com/introduction-to-mathematical-logic
RW-4	https://www.tutorialspoint.com/set-theory
RW-5	https://www.javatpoint.com/permutation-and-combinations

Video Links (VL):

VL-1	Introduction to Matrices https://youtu.be/T7I_C_IL75I
VL-2	Linear Equations - Homogeneous equations https://youtu.be/7a9Yawt1O50
VL-3	Non homogeneous equations https://youtu.be/iwkFNEWqzgc
VL-4	Mathematical Logics https://youtu.be/UM_i1Cs1Vzw
VL-5	Permutations and Combinations https://youtu.be/GZATQeI8dJo
VL-6	Set Theory https://youtu.be/tyDKR4FG3Yw

MOOC Courses (MC):

Sr.No.	Platform	Topic	CO	Link	Duration
MC-1	LinkedIn Learning	Every day math in life User – Offered by Google – Coursera.	C01	https://www.linkedin.com/learning/learning-everyday-math?trk=share_android_course_learning&shareId=JED2ywPrQeOX6GidIK5H5A%3D%3D	2 hours 6 min
MC	LinkedIn	Applied	CO	https://www.udemy.com/share/106E7e/	3

-2	n Learnin g	Mathemat ics - Mathemat ical reasoning	2 CO 3		hours 5 min
MC -3	Linkedl n Learnin g	Set Theory.	CO 3 CO 4	https://www.linkedin.com/learning/programming-foundations-discrete-mathematics?trk=share_android_course_learning&shareId=uxnlwE5%2FQDS%2By2UWCZZ7Sq%3D%3D	2 hours 15 min
MC -4	Charles Sturt Univers ity, Australi a	Introduct ion to permutat ion and Combin ation	CO5	https://www.linkedin.com/learning/excel-2007-business-statistics?trk=share_android_course_learning&shareId=EdbfsaBFS22Q93I%2BI0MSZQ%3D%3D	4 hour s 19 min
Total MOOC Course integration with Certification					12 hour s

Session-wise Planning:

Module	Session	Topic	TB/RB/OR/RW/VL	Pedagogy/ Activity planned	CO	Mode of delivery
Bride Course	Zero Lecture	Basic Mathematics	VL-1 & VL-5	Blended Learning Approach		Asynchronous-Video
1	2	Matrix Theory, Introduction, matrix definition with examples	TB-1, RB-1, RW-1	Open discussion session	CO1	Classroom Interaction
	2	types of matrix,		Open discussion session	CO1	Classroom Interaction
	2	operations on matrices.		Open discussion session	CO1	Classroom Interaction
	1	Elementary row and transformations		Interactive discussion	CO1	Classroom Interaction
	2	Inverse of a matrix by elementary transformation,		Interactive discussion	CO1	Classroom Interaction
	2	Rank of a matrix, Determination of rank of a matrix by reducing it to echelon form		Interactive discussion	CO1	Classroom Interaction
	1	Linear equations, definitions and examples.		Interactive discussion	CO1	Classroom Interaction
02	2	Linear equations introduction	TB-1, RB-1, RW-2	Interactive discussion	CO2	Classroom Interaction
	1	Homogeneous linear equations,		Interactive discussion	CO2	Classroom Interaction
	2	Working rule of Homogeneous linear equations,		Presentation and Interactive discussion	CO2	Classroom Interaction
	1	Non – homogenous equations.		Ppt presentation and	CO2	Classroom Interaction

				Interactive discussion		
	2	Working rule of Non – homogenous equations.		Ppt presentation and Interactive discussion	CO2	Classroom Interaction
	2	Trivial equations and problems		Ppt presentation and Interactive discussion	CO2	Classroom Interaction
	2	non trivial equations and problems		Ppt presentation and Interactive discussion	CO2	Classroom Interaction
03	2	Mathematical Logic : Open sentences	TB-1, RB-1, TB-2, RB-2, RW-3	Flipped class with MCQs and Interactive discussion	CO3	Classroom Interaction
	2	Compound open sentences		Flipped class with MCQs and Interactive discussion	CO3	Flip class
	2	quantifiers,		Interactive discussion	CO3	Flip class
	1	Truth tables		Interactive discussion	CO3	Classroom Interaction
	2	Connectives involving quantifiers		Interactive discussion	CO3	Classroom Interaction
	2	Logical implications		Assessment	CO3	Classroom Interaction
04	1	Set Theory : Fundamentals of set operations	TB-2, RB-1, RB-2, RW-4	Discussing the concept with examples	CO4	Classroom Interaction
	2	Subsets		Interactive discussion with Assessment	CO4	Flip class

	1	Counting and Venn Diagrams		Open discussion and problem solving	CO4	Classroom Interaction
	1	Relations		Open discussion and problem solving	CO4	Classroom Interaction
	1	Functions: Plain and one to one		Open discussion and problem solving	CO4	Classroom Interaction
	2	Onto Functions: Stirling Numbers of the second kind		Open discussion and problem solving	CO4	Classroom Interaction
	1	Special Functions		Open discussion and problem solving	CO4	Classroom Interaction
	1	The Pigeonhole Principle		Ppt presentation and Assessment	CO4	Flip class
	2	Function Composition		Open discussion and problem solving	CO4	Classroom Interaction
	1	Inverse Functions		Open discussion and problem solving	CO4	Classroom Interaction
05	1	Permutations and Combinations :Introduction	TB-2, RB-2, TB-3, RW-5	Ppt Presentation	CO5	Classroom Interaction
	2	The rule of sum and product		Ppt Presentation	CO5	Classroom Interaction
	2	Permutations		Ppt Presentation	CO5	Flip class

	2	Permutations with Repetitions		Ppt Presentation	CO5	Flip class
	2	Permutations of Objects not all Different		Ppt Presentation	CO5	Classroom Interaction
	2	Circular Permutations		Ppt Presentation	CO5	Classroom Interaction

Assessment Scheme: CA: UE – 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency/Stages	Weightage (%)	CO
1.	Skill Enhancement Activity-1 - Ppt presentation	Formative	1	15	CO1, CO2, CO3
2.	Content Improvement Activity-2 - Poster presentation		1/ 3	15	CO2, CO3, CO4, CO5
3.	Class Participation	Formative	-	5	-
4.	Periodic Test	Formative	3	15	CO1, CO2, CO3, CO4, CO5
5.	End Semester Examination	Summative	1	50	CO1, CO2, CO3, CO4, CO5
Total				100	

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